

- 2 The space race started in 1957, the year that the Soviet Union put the first artificial satellite, Sputnik 1 in orbit around the Earth. Sputnik 1 was launched using a rocket similar to the one shown in Fig. 2.1.

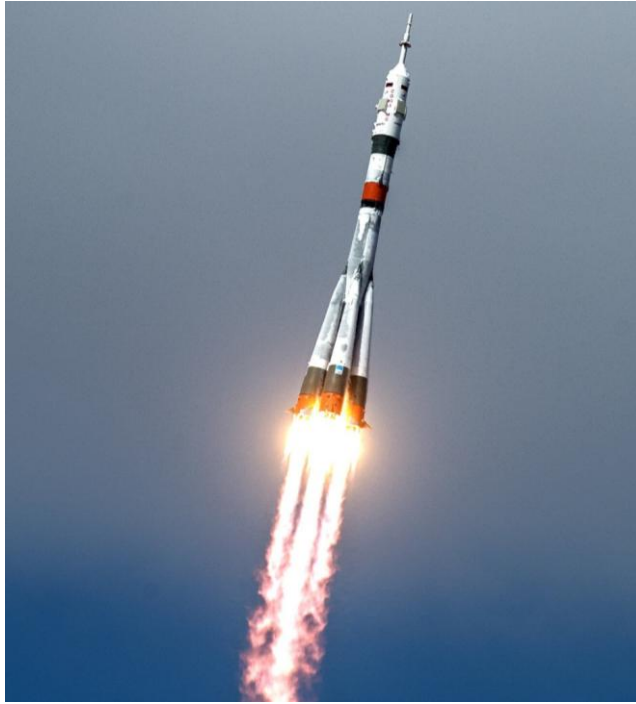


Fig. 2.1

The rocket was propelled by hot gases from the burning of a mixture of liquid oxygen and kerosene.

- (a) Using momentum consideration, explain how the ejection of hot gases can result in a forward thrust on the rocket.

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 [2]

- (b) Explain why the resultant force on the rocket in a vacuum is larger than when it is in air.

.....

 [2]

- (c) Fig. 2.2 shows a rocket in a vertical take-off position.

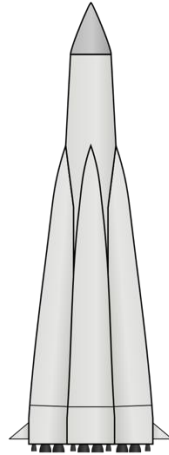


Fig. 2.2

The rocket, with its fuel, has a total mass of 1.2×10^6 kg.
Immediately after a vertical take-off, its acceleration is 3.5 m s^{-2} .

Assume that air resistance is negligible.

- (i) On Fig. 2.2, draw arrows to show the forces acting on the rocket immediately after take-off. Label the forces. [2]
- (ii) Determine the thrust of the rocket motors immediately after take-off.

thrust = N [2]

- (iii) 1.0 minute after the take-off, the rocket is still rising vertically. The thrust remains constant and its acceleration is 18.7 m s^{-2} .

Determine the average rate of fuel consumption during this 1.0 minute interval.

rate of fuel consumption = kg s^{-1} [2]

[Total: 10]

Turn over